

Gate access logic

How QR scans are evaluated at entrance and exit gates

Overview

This flow governs what happens when a child scans a QR code at either the entrance or exit gate of the facility. The system runs through a series of checks to decide whether to let the child through, and if so, what state to put them in (playing or on break). Every decision ends in exactly one of two outcomes: the scan is logged and the gate opens, or access is denied and nothing further happens — the gate never receives an open signal on a denial.

1. QR validation

The very first check is whether the scanned QR code is valid and tied to an active reservation. This is a hard gate: if the QR is invalid, expired, or has no reservation behind it, the flow ends immediately in **Access Denied**. No gate-type check, logging, or gate-open signal happens beyond this point. Only a valid QR with a reservation proceeds to the next step, where the system determines which gate is being used.

2. Which gate?

Once the QR is confirmed valid, the system checks whether the scan happened at the **entrance** or the **exit**. This single decision splits the flow into two independent branches with different rules, described below.

3. Entrance branch

The entrance branch exists to make sure a child is only let in if they are genuinely eligible to start or resume play. It runs three checks in sequence:

- **Finished playing / already left?** — If the child's record shows they already completed their visit or left for the day, entry is denied — a completed visit cannot be reopened by scanning again.
- **Currently playing?** — If the child is already marked as actively playing (they never checked out), a second entrance scan is treated as invalid and denied, since they should not be able to double up on a single session.
- **Currently on break?** — This is the deciding factor for children who pass both prior checks. If they are on a break, the system allows entry and **resumes** their play session. If they are not on a break (this is effectively their first entrance), the system allows entry and **starts** a new play session.

Both successful entrance outcomes (resume or start playing) converge on the same next step: logging the scan.

4. Exit branch

The exit branch makes sure only children with a legitimate, active check-in can leave for a break, and that break usage stays within the daily limit. It also runs three checks in sequence:

- **Checked in today?** — If there is no record of the child checking in at all today, the exit scan is denied — there is nothing valid to exit from.

- **Finished playing / already left?** — If the child has already been marked as finished or has already left, a further exit scan is redundant and denied.
- **Taken 5 breaks?** — This enforces the daily break limit. If the child has already used all 5 allowed breaks, exit is denied to prevent additional breaks. Otherwise, exit is allowed and a new break is started for the child.

5. Logging and gate control

Every successful outcome — resuming play, starting play, or starting a break — is routed to a shared **Log the scan** step, which records the event before anything physically happens at the gate. Only after logging does the system **tell the gate to open**.

Access Denied is a dead end by design: it does not log a scan and it does not trigger the gate-open signal, regardless of which check caused the denial. This keeps the gate closed to anyone who fails validation at any point in the flow.

6. Denial conditions at a glance

Gate	Denial condition	Reason
Any	Invalid QR / no reservation	No valid entry to begin with
Entrance	Already finished / left	Visit already completed
Entrance	Already marked as playing	Prevents duplicate active sessions
Exit	Not checked in today	Nothing valid to exit from
Exit	Already finished / left	Redundant exit
Exit	5 breaks already taken	Enforces daily break limit

Flowchart

